Relative entropy in CFT

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1 Setup

We follow Nima's paper 1508.03506 to compute relative entropies in CFT.

Relative entropy is defined as

$$S(\phi\|\psi) = \operatorname{tr}\phi\log\phi - \operatorname{tr}\phi\log\psi \tag{1}$$

for two density matrices ϕ, ψ. The above formula assumes the density matrices are normalized, but we will not assume this in later calculation. The relative entropy can be written as a limit

$$S(\phi\|\psi) = \partial_n\left(\frac{\operatorname{tr}(\phi^n)}{\operatorname{tr}(\phi)^n} - \frac{\operatorname{tr}(\phi\psi^{n-1})}{\operatorname{tr}(\phi)\operatorname{tr}(\psi)^{n-1}}\right)\Bigg|_{n\to 1} \tag{2}$$

$$= \frac{\partial_n(\operatorname{tr}(\phi^n)/\operatorname{tr}(\phi)^n)}{\operatorname{tr}(\phi^n)/\operatorname{tr}(\phi)^n} - \frac{\partial_n(\operatorname{tr}(\phi\psi^{n-1})/\operatorname{tr}(\phi)\operatorname{tr}(\psi)^{n-1})}{(\operatorname{tr}(\phi\psi^{n-1})/\operatorname{tr}(\phi)\operatorname{tr}(\psi)^{n-1})}\Bigg|_{n\to 1} \tag{3}$$

in the second line we added the denominator since they go to 1 as $n \to 1$. This gives the main formula that we will use:

$$S(\phi\|\psi) = \partial_n \log\left[\frac{\operatorname{tr}(\phi^n)\operatorname{tr}(\psi)^{n-1}}{\operatorname{tr}(\phi\psi^{n-1})\operatorname{tr}(\phi)^{n-1}}\right]_{n\to 1} \tag{4}$$

We are interested in CFT defined on a cylinder, where the circle size is R. We take an interval $(-l/2, l/2)$ of size l, and define the ratio $x \equiv l/R$.

In the above formula, each term can be evaluated in using a path integral on a replicated manifold. See figures in Nima's paper.

$$S(\phi||\psi) = \partial_n \log\left[\frac{\langle\mathcal{O}_\Phi^{(n)}\rangle_{\mathcal{R}_n}\langle\Psi(z_1')\Psi(z_1)\rangle_{\mathcal{R}_1}^{n-1}}{\langle\Phi(z_n')\Phi(z_n)\mathcal{O}_\Psi^{(n-1)}\rangle_{\mathcal{R}_n}\langle\Phi(z_1')\Phi(z_1)\rangle_{\mathcal{R}_1}^{n-1}}\right]_{n\to 1} \tag{5}$$

Let's say we have a complex coordinate z on the cylinder. States (bras/kets) are created from operators inserted at $z = \pm i\infty$ on each sheet. The interval is at $t = \operatorname{Im} z = 0$, and $\operatorname{Re} z \in (-l/2, l/2)$. We first transform it to the complex plane with coordinate w

$$w = \left(\frac{\sin(\pi(z+l/2)/R)}{\sin(\pi(z-l/2)/R)}\right)^{1/n}. \tag{6}$$

Now the operators are mapped to $w = \exp(2\pi ik/n \pm \pi ix/n)$, $k = 0, ..., n-1$. The intervals on each sheet are mapped to the half-infinite lines $w = \exp(2k\pi i/n)$, while the complement regions are mapped to half-infinite lines $w = \exp((2k+1)\pi i/n)$.

Then we do another conformal transformation

$$\omega = e^{it} \tag{7}$$

to go to a strip with width 2π. Now the operator insertions are at $t = 2\pi k/n \pm \pi x/n$, and it should be kept in mind that the edges $t = -\pi, \pi$ are identified. All the later calculations will be on the strip, and Nima showed that the difference caused by the conformal transformation goes to zero when $n \to 1$.

2 Free boson correlators

In free boson CFT, primary operators are vertex operators

$$V_\alpha =: e^{i\alpha\phi} : \tag{8}$$

And there correlators are given by

$$\left\langle \prod_k V_{a_k}(t_k) \right\rangle = \prod_{k>i} \left[2\sin\left(\frac{t_{ki}}{2}\right)\right]^{a_k a_i} \tag{9}$$

on the right, the product is over all possible Wick contractions.

Let's first reproduce Nima's answer for $S(\alpha||\beta)$. To compute this, we essentially need the following correlator:

$$\langle V_{-\alpha}\mathcal{O}_\beta^{(n-1)}V_\alpha\rangle_S = (2\sin(\pi x/n))^{-\beta^2(n-1)-\alpha^2}\left(\frac{\sin(\pi x)}{n\sin(\pi x/n)}\right)^{-(n-2)\beta^2-2\alpha\beta} \tag{10}$$

where $O_\beta^{(n-1)}$ means $(n-1)$ pairs of $V_\beta V_{-\beta}$ multiplied together. And,

$$\langle O_\alpha^{(n)}\rangle = (2\sin(\pi x/n))^{-n\alpha^2}\left(\frac{\sin(\pi x)}{n\sin(\pi x/n)}\right)^{-n\alpha^2} = \left(\frac{2}{n}\sin\pi x\right)^{-n\alpha^2} \tag{11}$$

From this,

$$S(\alpha\|\beta) = \partial_n \log\left(\frac{\langle\mathcal{O}_\alpha^{(n)}\rangle_S \langle V_{-\beta}V_\beta\rangle_S^{n-1}}{\langle V_{-\alpha}V_\alpha\mathcal{O}_\beta^{(n-1)}\rangle_S \langle V_{-\alpha}V_\alpha\rangle_S^{n-1}}\right) = (\alpha-\beta)^2(1-\pi x\cot(\pi x)). \tag{12}$$

To derive the first correlator, let us find all the Wick contractions. We assume the two α's are located at $\pm\pi x/n$. Here are all the contractions, and we need to multiply them together:

- $(\alpha, -\alpha)$: This gives

$$\left(2\sin\frac{\pi x}{n}\right)^{-\alpha^2} \tag{13}$$

- $(\beta, -\beta)$ for nearest neighbors.

$$\left(2\sin\frac{\pi x}{n}\right)^{-(n-1)\beta^2} \tag{14}$$

- $(\beta, -\beta)$ for non-nearest neighbors

$$\prod_{k=1}^{n-2}(2\sin((k+x)\pi/n))^{(n-k-1)\beta^2}(2\sin((k-x)\pi/n))^{(n-k-1)\beta^2} \tag{15}$$

- (β,β) and $(-\beta,-\beta)$

$$\prod_{k=1}^{n-2}\left(2\sin(k\pi/n)\right)^{-2(n-k-1)\beta^2} \tag{16}$$

- (α,β) and $(-\alpha,-\beta)$

$$\prod_{k=1}^{n-1}(2\sin(k\pi/n))^{2\alpha\beta}=n^{2\alpha\beta} \tag{17}$$

- $(\alpha,-\beta)$ and $(-\alpha,\beta)$

$$\prod_{k=1}^{n-1}(2\sin((k+x)\pi/n))^{-\alpha\beta}\prod_{k=1}^{n-1}(2\sin((k-x)\pi/n))^{-\alpha\beta} \tag{18}$$

To get the final answer, we need to use the product formulas in Nima's appendix.

For $\langle O_\alpha^{(n)}\rangle$, we only need different contractions between α's and $-\alpha$'s to get the answer.

3 Relative entropy for mixed states

Now we want to compute

$$S(\rho||(\rho+\sigma)/2) \tag{19}$$

which is

$$S(\rho\|\frac{\rho+\sigma}{2})=\partial_n\log\left[\frac{\mathrm{tr}\,(\rho^n)\,[\mathrm{tr}(\rho+\sigma)]^{n-1}}{\mathrm{tr}\,(\rho(\rho+\sigma)^{n-1})\,[\mathrm{tr}(\rho)]^{n-1}}\right]_{n\to1} \tag{20}$$

So we need

$$\mathrm{tr}(\rho(\rho+\sigma)^{n-1})=\sum_{k=0}^{n-1}\binom{n-1}{k}\mathrm{tr}(\rho^{n-k}\sigma^k)+\text{terms with commutators} \tag{21}$$

For free boson CFT correlators, the order of operators does not matter, so the terms with commutators will vanish.

To be more general, let us compute

$$\langle O_\alpha^{(p)}O_\beta^{(q)}\rangle=\langle V_{-\alpha}V_\alpha...V_{-\alpha}V_\alpha V_{-\beta}V_\beta...V_{-\beta}V_\beta\rangle \tag{22}$$

where there are p pairs of V_α's and q pairs of V_β's.

The final answer is

$$\langle O_\alpha^{(p)}O_\beta^{(q)}\rangle=(2\sin(\pi x/n))^{-p\alpha^2-q\beta^2}g(p,q;x)^{\alpha^2}g(q,p;x)^{\beta^2}h(p,q;x)^{\alpha\beta} \tag{23}$$

where

$$g(p,q;x)\equiv\prod_{k=1}^{p-1}\left(\frac{\sin^2\frac{k\pi}{p+q}}{\sin\frac{(k+x)\pi}{p+q}\sin\frac{(k-x)\pi}{p+q}}\right)^{p-k}=??? \tag{24}$$

and

$$h(p,q;x)=\prod_{\ell=1}^{q}\prod_{k=1}^{p}\frac{\sin^2((k+\ell-1)\pi/n)}{\sin((k+\ell-1+x)\pi/n)\sin((k+\ell-1-x)\pi/n)}=??? \tag{25}$$

and we don't know their closed forms. Without closed forms, it will be hard to get the final answer.

Here is the derivation from all the Wick contractions ($p+q=n$):

- $(\alpha, -\alpha)$ for nearest neighbors

$$\left(2\sin\frac{\pi x}{n}\right)^{-p\alpha^2} \tag{26}$$

- $(\alpha, -\alpha)$ for non-nearest neighbors

$$\prod_{k=1}^{p-1}[(2\sin((k+x)\pi/n))\,(2\sin((k-x)\pi/n))]^{(p-k)\alpha^2} \tag{27}$$

- (α, α) and $(-\alpha, -\alpha)$

$$\prod_{k=1}^{p-1}(2\sin(k\pi/n))^{-2(p-k)\alpha^2} \tag{28}$$

- $(\beta, -\beta)$ for nearest neighbors

$$\left(2\sin\frac{\pi x}{n}\right)^{-q\beta^2} \tag{29}$$

- $(\beta, -\beta)$ for non-nearest neighbors

$$\prod_{k=1}^{q-1}[(2\sin((k+x)\pi/n))\,(2\sin((k-x)\pi/n))]^{(q-k)\beta^2} \tag{30}$$

- (β, β) and $(-\beta, -\beta)$

$$\prod_{k=1}^{p-1}(2\sin(k\pi/n))^{-2(q-k)\beta^2} \tag{31}$$

- (α, β) and $(-\alpha, -\beta)$

$$\prod_{\ell=1}^{q}\prod_{k=1}^{p}(2\sin((k+\ell-1)\pi/n))^{2\alpha\beta} \tag{32}$$

- $(\alpha, -\beta)$ and $(-\alpha, \beta)$

$$\prod_{\ell=1}^{q}\prod_{k=1}^{p}(2\sin((k+\ell-1+x)\pi/n))^{-\alpha\beta}\prod_{\ell=1}^{q}\prod_{k=1}^{p}(2\sin((k+\ell-1-x)\pi/n))^{-\alpha\beta} \tag{33}$$

For the unknown math formula:

$$g(p,q;x) \equiv \prod_{k=1}^{p-1}\left(\frac{\sin^2\frac{k\pi}{p+q}}{\sin\frac{(k+x)\pi}{p+q}\sin\frac{(k-x)\pi}{p+q}}\right)^{p-k} = ??? \tag{34}$$

We only know it for $q = 0, 1$. When $q = 0$, it is

$$\left(\frac{p\sin(\pi x/p)}{\sin(\pi x)}\right)^{p} \tag{35}$$

When $q = 1$, it is

$$\left(\frac{(p+1)\sin(\pi x/(p+1))}{\sin(\pi x)}\right)^{p-1} \tag{36}$$

The obvious generalization of this formula turns out to not be true.

We also need

$$h(p,q,x) \equiv \prod_{\ell=1}^{q}\prod_{k=1}^{p}\frac{\sin^2((k+\ell-1)\pi/n)}{\sin((k+\ell-1+x)\pi/n)\sin((k+\ell-1-x)\pi/n)} = ??? \tag{37}$$

4 The limit of small intervals

Due to the interest of approximate quantum error correcting code, we are interested in the case the $S^1 = AB$ with two intervals A and B, in the limit $x_A \to 0$, where $x_A \equiv |A|/L$ and compute

1. (small interval quantity) $S(\rho_A^a || \bar{\rho}_A)$
2. (large interval quantity) $S(\rho_B^a || \bar{\rho}_B)$

where $\bar{\rho} = \frac{1}{N} \sum_a \rho^a$.

4.1 Free Boson RCFT

In the context of free boson RCFT ($c = 1$) at $R = \sqrt{2k}$, we can take $\bar{\rho} = \frac{1}{2}(\rho^1 + \rho^s)$ with a certain Virasoro primary $|s, I, J\rangle$[1] associated with an Abelian anyon s such that $s \times s = 1$ and let $a = 1, s$. Based on physical intuition, we should have

$$S(\rho_A^a || \bar{\rho}_A) \sim x_A^\alpha, \tag{38}$$

at the limit of $x_A \to 0$. For large interval B, the complement of A, we should have

$$\log N - S(\rho_B^a || \bar{\rho}_B) \sim x_A^\beta, \tag{39}$$

to the leading powers of x_A. The leading power is the smallest positive power. It is important to compute the powers α and β given the CFT and the selected primary states.

Can the replica trick help? The value β is computed using small size numerics. Let us make an attempt to compute β using the replica trick.

Conjecture: For free boson $S(\rho_B^1 || \bar{\rho}_B) = S(\rho_B^s || \bar{\rho}_B)$, where $\bar{\rho} = (\rho^1 + \rho^s)/2$.

ν	α	β
1/2	2.0	1.149
1/4	2.0	0.762

Table 1: Powers of α, β of relative entropy for RCFT at $R = \sqrt{1/\nu}$. α is from references [] is verified numerically. β is from finite sizes.

Near term goal: use replica trick to predict β independently, and to contrast the values in Table 1. In order to do so, we should compute β_n according to

$$x_A^{\beta_n} \sim \log 2 - S^n(\rho_B^a || \bar{\rho}_B). \tag{40}$$

4.2 Analytic formulas

This is the analytic result from Nima:

$$S\left(\rho_a \| y\rho_a + (1-y)\rho_b\right) = -\log\left(y + (1-y)\exp\left[-(a-b)^2(1 - \pi x \cot(\pi x))\right]\right) \tag{41}$$

where a (or b) is the conformal weight of the vertex operator $: e^{ia\phi} :$. We are interested in the case where $y = 1/2$.

[1] To specify a Virasoro primary, we need to choose I and J and let $|\psi^s\rangle = |s, I, J\rangle$; see the main text for the details. The vacuum state corresponds to $|\psi^1\rangle = |1, 0, 0\rangle$.

Expanding the above formula for $x \ll 1$, we get

$$S(\rho_a || \frac{\rho_a + \rho_b}{2}) = \frac{\pi^2}{6}(a-b)^2 x^2 + O(x^4). \tag{42}$$

Therefore $\alpha = 2$. Expanding near $x \sim 1$, we get

$$S(\rho_a || \frac{\rho_a + \rho_b}{2}) = \log(2) - e^{-\frac{(a-b)^2}{1-x}} \left(1 + \frac{\pi^2}{3}(a-b)^2(1-x) + O((1-x)^2)\right), \tag{43}$$

which is not a power law decay as expected. We should understand this better. This formula does not fit our data, as in Fig. 2 and in Table 1.

4.3 Compare with the numerical data of Coherent Information Drop

The coherent information drop induced by erasing area Q_A is given by

$$\Delta(Q_A) = \log D + \frac{1}{D}\sum_i \left(S(\rho^i_{Q_A} || \bar{\rho}_{Q_A}) - S(\rho^i_{\bar{Q}_A} || \bar{\rho}_{\bar{Q}_A})\right), \tag{44}$$

as is explained in the main text. Let Q_A be a single interval on the circle and set $x = |Q_A|/L$. By Nima's formula Eq. (41), we have (writing $\Delta(Q_A)$ as $\Delta(x)$ for short)

$$\begin{aligned} \Delta(x) = \log 2 - \log\Big(\frac{1}{2} + \frac{1}{2}\exp\big(-(a-b)^2(1 - \pi x \cot(\pi x))\big)\Big) \\ + \log\Big(\frac{1}{2} + \frac{1}{2}\exp\big(-(a-b)^2(1 - \pi(1-x)\cot(\pi(1-x)))\big)\Big). \end{aligned} \tag{45}$$

4.4 Compare with the numerical data of Relative entropy with maximally-mixed state

Using the joint convexity of relative entropy, we have

$$\begin{aligned} S(\rho^{00}_A || \bar{\rho}_A) &\leq \frac{1}{2} S(\rho^{00}_A || \rho^{st}_A) \\ &= \frac{1}{4}(1 - \pi x \cot \pi x) \end{aligned} \tag{46}$$

as a *upper bound*. The first line follows from the joint convexity of the relative entropy. The second line follows from Nima's formula for the relative entropy between two primary states [1, 2].

Let us contrast this with Nima's new formula Eq. (41). Nima's new analytical results reads

$$S(\rho^{00}_A || \bar{\rho}_A) = -\log\left(\frac{1}{2} + \frac{1}{2}\exp\left[-\frac{1}{2}(1 - \pi x \cot(\pi x))\right]\right) \tag{47}$$

We plot the upper bound, analytical results and the numerical results together and found their consistency for $x \ll 1$.

- ($x \ll 1$) The numerical test shows good agreement between Nima's new analytical formula (47), joint convexity upper bound (46) and the numerical data at $x \ll 1$. This is shown in Fig. 1.

- $(1-x \ll 1)$ The coherent information loss (45) cannot be characterized well by the new result, as shown in Fig. 2. This appear to suggest that Nima's new formula [Eq. (41)] does not fit the data for $1-x \ll 1$ (i.e. $x \to 1$).

  The value of β in Table 1 may give a hint on improvement.

  Note that, the reason we numerically compute $\Delta(x)$ for small x instead of computing relative entropy for large x is due to the numerical efficiency.

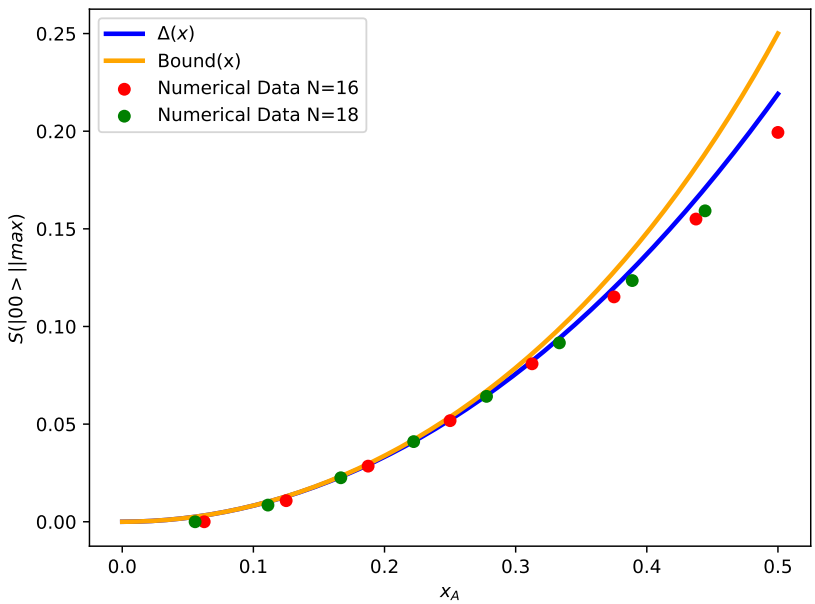

Figure 1: Relative entropy $S(\rho_A^{00}||\bar{\rho}_A)$ vs $x = |A|/L$ of $R = \sqrt{2}$ RCFT with $c = 1$ (from $\nu = 1/2$ Laughlin state of Ref. [3,4],) with code space $\{|00\rangle, |st\rangle\}$, which is created by the vertex operators with $b = 0$ and $a = \frac{1}{\sqrt{2}}$. The numerical results are from system size $N = 16, 18$ on the circle.

A Coherent information loss versus relative entropy

Let the entire system by Q, and let $Q = Q_1 Q_2$. We have shown in the main text that, the coherent information loss due to erasure of region Q_1 is

$$\Delta(Q_1) = (S_{Q_1} + S_Q - S_{Q_2})_{\bar{\rho}_Q}, \quad \bar{\rho}_Q = \frac{1}{N}\sum_{i=1}^{N} |\psi_Q^i\rangle\langle\psi_Q^i|, \tag{48}$$

where $\{|\psi_Q^i\rangle\}_{i=1}^N$ is an orthonormal set of states.

It is possible to show

$$\boxed{\Delta(Q_1) = \log N + \frac{1}{N}\sum_{i=1}^{N}\left(S(\rho_{Q_1}^i||\bar{\rho}_{Q_1}) - S(\rho_{Q_2}^i||\bar{\rho}_{Q_2})\right).} \tag{49}$$

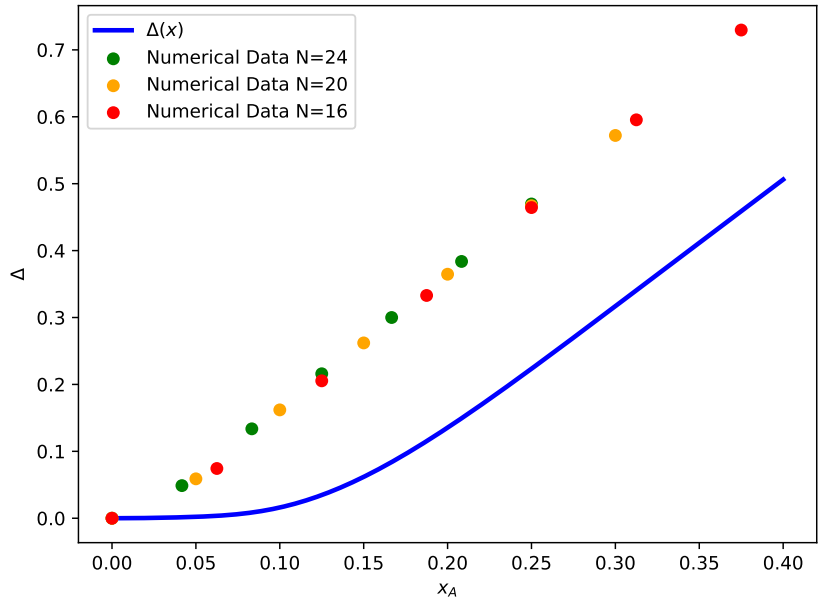

Figure 2: Coherent information loss Δ vs x of $R = \sqrt{2}$ RCFT with missing qubit error that knocks off interval A. The code space is $\{|00\rangle, |st\rangle\}$, spanned by two Virasoro primaries: the vacuum with $(h_L, h_R) = (0, 0)$ and another one with $(h_L, h_R) = (1/4, 1/4)$. They created by the vertex operators with $b = 0$ and $a = \frac{1}{\sqrt{2}}$. The numerical results are from $N = 16, 20, 24$ on the circle.

Below is the proof. First, $S_Q = \log N$. Second,

$$\begin{aligned} & S_{Q_1}(\bar{\rho}_Q) - S_{Q_2}(\bar{\rho}_Q) \\ & = \frac{1}{N} \sum_{i=1}^{N} \left(S(\rho^i_{Q_1} || \bar{\rho}_{Q_1}) - S(\rho^i_{Q_2} || \bar{\rho}_{Q_2}) \right) . \end{aligned} \tag{50}$$

References

[1] N. Lashkari, "Relative Entropies in Conformal Field Theory," *Phys. Rev. Lett.* **113** (Aug., 2014) 051602, `1404.3216`.

[2] N. Lashkari, "Modular Hamiltonian for Excited States in Conformal Field Theory," *Phys. Rev. Lett.* **117** (July, 2016) 041601, `1508.03506`.

[3] A. E. B. Nielsen, J. I. Cirac, and G. Sierra, "Laughlin Spin-Liquid States on Lattices Obtained from Conformal Field Theory," *Phys. Rev. Lett.* **108** (June, 2012) 257206, `1201.3096`.

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